Instructions:

- Clearly explain all your answers.
- For each True/False question, state your answer and provide a brief justification.
- Make necessary assumptions and clearly state all the assumptions.

Submission Date: August 20, 2025, 5:00 PM.

Problem 1

Let V_1, V_2 be subspaces of a vector space V. Which of the following is necessarily a subspace of V?

(a) $V_1 \cap V_2$

(b) $V_1 \cup V_2$

(c) $V_1 + V_2 = \{x + y : x \in V_1,\ y \in V_2\}$

(d) $V_1 \setminus V_2 = \{x \in V_1 \text{ and } x \notin V_2\}$

Problem 2

Select all choices that are subspaces of $\mathbb{R}^3$.

Note: $\mathbb{R}$ denotes the set of real numbers.

(a) $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},\ \alpha, \beta \in \mathbb{R} \right\}$

(b) $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : \mathbf{x} = \alpha^2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \beta^2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},\ \alpha, \beta \in \mathbb{R} \right\}$

(c) $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 = 0, \quad 4x_1 - 2x_2 + 3x_3 = 0 \right\}$

(d) $\left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : 5x_1 + 2x_3 + 4 = 0 \right\}$

Problem 3

Which of the following statements is/are TRUE?

Note: $\mathbb{R}$ denotes the set of real numbers.

(a) There exist $M \in \mathbb{R}^{3\times 3}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has a unique solution and $Mx = q$ has infinite solutions.

(b) There exist $M \in \mathbb{R}^{3\times 3}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has no solutions and $Mx = q$ has infinite solutions.

(c) There exist $M \in \mathbb{R}^{2\times 3}$, $p \in \mathbb{R}^2$, and $q \in \mathbb{R}^2$ such that $Mx = p$ has a unique solution and $Mx = q$ has infinite solutions.

(d) There exist $M \in \mathbb{R}^{3\times 2}$, $p \in \mathbb{R}^3$, and $q \in \mathbb{R}^3$ such that $Mx = p$ has a unique solution and $Mx = q$ has no solutions.

Problem 4

Consider $\mathbb{R}^3$ as a vector space with the usual operations of vector addition and scalar multiplication. Let $x \in \mathbb{R}^3$ be denoted by $x = (x_1, x_2, x_3)$. Define subspaces W_1 and W_2 by

$$W_1 := \{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 0\}$$

and

$$W_2 := \{x \in \mathbb{R}^3 : 2x_1 + 3x_3 = 0\}.$$

Let $\dim(U)$ denote the dimension of the subspace U.

Which of the following statements are TRUE?

(a) $\dim(W_1) = \dim(W_2)$

(b) $\dim(W_1) + \dim(W_2) - \dim(\mathbb{R}^3) = 1$

(c) $\dim(W_1 + W_2) = 2$

(d) $\dim(W_1 \cap W_2) = 1$

Problem 5

What two elimination matrices E_{21}, E_{32} put A into its upper triangular form $E_{32}E_{21}A = U$?

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}$$

Problem 6

Find the inverse of the following matrices and also validate your answer.

(a)

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

(b)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

Problem 7

Solve the system of linear equations by first decomposing A into LU, then solve $Ax = b$ using LU. Do you feel any difference or benefit in using LU decomposition?

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 4 \\ -6 \\ 7 \end{bmatrix}$$

Problem 8

Find the rank of the following matrix A:

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 1 \\ 2 & 4 & 1 & 3 & 0 \\ 3 & 6 & 2 & 5 & 1 \\ -4 & -8 & 1 & -3 & 1 \end{bmatrix}$$

Problem 9

Let R be the row reduced echelon form of a 4×4 real matrix A and let the third column of R be $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$. Consider the following statements:

P: If $\begin{bmatrix} \alpha \\ \beta \\ \gamma \\ 0 \end{bmatrix}$ is a solution of $Ax = 0$, then $\gamma = 0$.

Q: For all $b \in \mathbb{R}^4$, $\text{rank}[A \,|\, b] = \text{rank}[R \,|\, b]$.

Then

(a) both P and Q are TRUE

(b) P is TRUE and Q is FALSE

(c) P is FALSE and Q is TRUE

(d) both P and Q are FALSE

Problem 10

For a matrix $M = [m_{ij}]$, $i, j = 1, 2, 3, 4$, the diagonal elements are all zero and $m_{ij} = -m_{ji}$. The minimum number of elements required to fully specify the matrix is ____.

(a) 0

(b) 6

(c) 12

(d) 16